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An Explainable Deep Learning Framework for Multi-Class Alzheimer's Diagnosis with Multi-Scale CNN

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***Abstract*—Early detection and accurate staging of Alzheimer's disease (AD) from structural MRI scans are crucial for timely clinical intervention. However, many existing deep learning approaches rely on conventional CNN architectures, are sensitive to evaluation design choices, and provide limited interpretability or deployment considerations. To address these challenges, this work presents a multi-scale, custom-designed Convolutional Neural Network (CNN) integrated with a MobileNetV2-based preprocessing stem and progressive convolutional blocks to capture structural MRI features at multiple receptive fields. Spatial feature consolidation with reduced computational overhead is achieved through a dual-path pooling mechanism and a Global Average Pooling–based classification head. An ensemble combining the proposed CNN with Xception and ResNet50 by majority vote is introduced to enhance prediction robustness for four AD stages: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. Strong multiclass performance was achieved when experiments were conducted on the Alzheimer's Disease Augmented MRI dataset, achieving an average accuracy of 99.15% $\pm$ 0.04 and an F1-score of 0.9915 $\pm$ 0.0004 under a 3-fold stratified cross-validation protocol, indicating high predictive performance with low variance across folds. So as to improve model transparency, discriminative neuroanatomical regions associated with disease severity have been visualized through the use of Grad-CAM-guided backpropagation, Integrated Gradients, and LIME. Moreover, the proposed framework has been deployed as a lightweight web application and made publicly available to showcase the real-time inference capability. The results indicate that the proposed approach offers an accurate and interpretable framework for multi-class AD staging and conveys promising potential for future clinical screening research.**

***Index Terms*—Alzheimer's Disease, Deep Learning, Convolutional Neural Network (CNN), MRI classification, Ensemble Learning, Explainable AI.**

I. Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disease and the most common cause of dementia that destroys essential mental faculties like memory, rationality, and behavior. Being one of the world's major health issues, AD has a significant worldwide prevalence. While the World Health Organization (WHO) estimates that 60–70% of the 55 million dementia cases are cases of AD [1], estimated cases worldwide are expected to hit 100 million by the year 2050 [2]. This makes it very necessary for us to have instruments that can assist the clinicians in managing the increasing population of patients.

Neuroimaging, and structural MRI in particular, is at the center of the clinical working algorithm for AD. It provides hallmark features like medial temporal lobe atrophy, hippocampal shrinkage, and decreased cortical thickness that are utilized for disease staging [3]. The radiological evaluation of these images involves discrimination between multiple levels of intellectual decline, from non-dementia to mid-degree dementia. The process, however, might be laborious and prone to inter-observer variability, thus calling for automatic systems that might streamline and expedite diagnosis [4]. Consequently, there is an increasing demand for reliable computational tools that can assist clinicians in achieving faster and more objective diagnostic decisions.

For these needs to be achieved, machine learning and deep learning methods have been widely used in neuroimaging data with the support of large public data collections such as ADNI and OASIS [5]. Convolutional Neural Networks (CNN) have demonstrated strong performance in medical image classification tasks; however, many strong models are computationally expensive and are essentially black boxes, limiting interpretability [6]. The challenge, therefore, is not just that of building accurate models, but of creating efficient, effective, and above all, understandable systems that can distinguish AD across its spectrum of severity. Clinicians must understand why a model makes a given prediction in order to use it as part of their trust in a diagnosis [7].

Motivated by these requirements, this paper presents a systematic deep learning framework for multi-class Alzheimer's disease severity classification. We design a computationally efficient custom CNN, an ensemble model to enhance robustness, and conduct comprehensive comparisons with state-of-the-art architectures. In addition, multiple explainable AI (XAI) techniques are integrated to elucidate model decision-making. Finally, the models are deployed in a user-friendly web application and the complete codebase is released, providing a transparent and deployable solution for automated AD severity assessment. The remainder of this paper is organized as follows. Section II reviews the related work. Section III presents the methodology. Section IV details the proposed model architecture. Section V reports the experimental results. Section VI provides a comparative analysis. Finally, Section VII concludes the paper and outlines future work.

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II. Related Work

AD classification from structural MRI has evolved from hand-crafted feature–driven pipelines to end-to-end deep learning architectures. Despite this progress, significant gaps persist in developing sufficiently robust and interpretable systems for reliable multi-stage clinical diagnosis.

Early methods for AD classification mainly used traditional machine learning that was trained based on hand-crafted features from manually delineated regions of interest. For example, Liu et al. [8] proposed the Whole Brain Hierarchical Network (WBHN). WBHN adopted multiscale atlas-based texture features along with inter-regional correlations, which resulted in outstanding performances at 94.65% for binary diagnostic tasks between AD, MCI, and healthy controls. However, such methods inherently rely on pre-defined feature engineering and are seldom validated for fine-grained four-stage disease severity classification, hence limiting their clinical value.

It was not until the breakthrough in deep learning, especially CNNs, that it became possible to perform automatic feature learning from raw MRI scans themselves. This marked a new shift in AD classification research. Though many CNN-based works have reported high accuracy, their performance was highly sensitive to experimental design choices. For instance, Bäckström et al. [9] showed that a 3D CNN achieved 98.74% accuracy for AD versus normal controls but decreased to 90.11% under a rigorous subject-independent splitting protocol. Also, this method needed extensive preprocessing. These ultimately reflected common pitfalls faced in the current literature: data leakage, limited generalizability, and strong reliance on preprocessing strategies. Subsequent studies by Kamath et al. [10] and Singh et al. [11] reinforced the idea of CNN-based architectures, although these studies were limited to binary classification and did not consider systematic interpretability analysis or deployment-oriented evaluation.

More recent initiatives have covered multiclass MRI-based AD staging and have obtained some encouraging results. Murugan et al. [12] proposed DEMNET, a custom CNN trained from scratch for four-class dementia classification using SMOTE-based class balancing and occlusion sensitivity maps, which reports strong performance on benchmark datasets. Rahman et al. [13] have performed a comparative analysis of several CNN architectures based on transfer learning and reported high classification accuracy on the same task. Along similar lines, Das et al. [14] have carried out a comparative study of several pre-trained CNN backbones such as ResNet50, InceptionV3, and EfficientNetB0 on the OASIS dataset, where EfficientNetB0 resulted in the highest validation accuracy after SMOTE-based balancing, showing the effectiveness of transfer learning for multiclass AD staging.

Other multiclass methods have furthered this line of research. For instance, Sharif et al. [15] introduced a CNN-based approach able to identify early and late MCI stages using unsupervised pretraining and visualizations from Grad-CAM. Farooq et al. [16] showed very high multiclass accuracy using very deep CNNs. However, these approaches mostly implement image-level splitting and do not consider the challenges in multi-scale representation learning, holistic explainability, and deployment-oriented assessment together, which is very important for reliable clinical translation.

In parallel, increasing computational efficiency and interpretability has received increased attention. Imtiaz et al. [17] reported a multi-class classification accuracy of 99.74% using their own CNN on the OASIS dataset; however, no more than a single imbalanced dataset was used for its assessment, with cross-dataset or subject-wise validation lacking, which made generalization hard to estimate. Another emerging direction is transformer-based architectures. Ahamed et al. [18] investigated Vision Transformer and Swin Transformer models on MRI scans, ensuring high multiclass results with extensive data augmentation. However, integrated explainability mechanisms were not considered, and robustness analysis beyond single-dataset evaluation was also missing, leaving the interpretability and generalization concerns insufficiently addressed.

Despite strong within-dataset performance of modern deep learning methods for Alzheimer's disease classification, several limitations continue to hinder clinical translation, including class imbalance and evaluation bias, a historical focus on binary detection rather than clinically meaningful multiclass staging, and limited consideration of deployment feasibility and integrated explainability. To address these gaps, this paper presents a comprehensive four-class AD severity classification framework that combines a custom multi-scale CNN, an ensemble strategy for robustness, explicit class-wise evaluation, complementary explainability methods, and lightweight web-based deployment to enhance transparency and practical applicability.

III. Methodology

A. Data Collection and Description

The dataset used in the experiment is the Alzheimer Disease Augmented MRI Dataset from Kaggle composed of the MRI brain images categorized into four stages of the disease : Mild Demented, Moderate Demented, Non Demented, and Very Mild Demented. The images are provided in the JPG format with the resolution of the images at 200×190 pixels.

B. Data Segmentation

The Kaggle Alzheimer's MRI dataset used in this study consists of two subsets: (1) an augmented dataset containing 33,984 MRI images across four classes, and (2) an unaugmented dataset comprising 5,099 original MRI images.

The augmented dataset was used for model development with the help of a three-fold stratified cross-validation process. In all the folds, two-thirds of the data was used for training, and one-third was used for validation. To handle the class imbalance, data augmentation was performed on the training data, and the validation data was not altered. The cross-validation score provided is the mean and standard deviation of the three folds. Cross-validation performance is reported as mean $\pm$ standard deviation across the three folds.

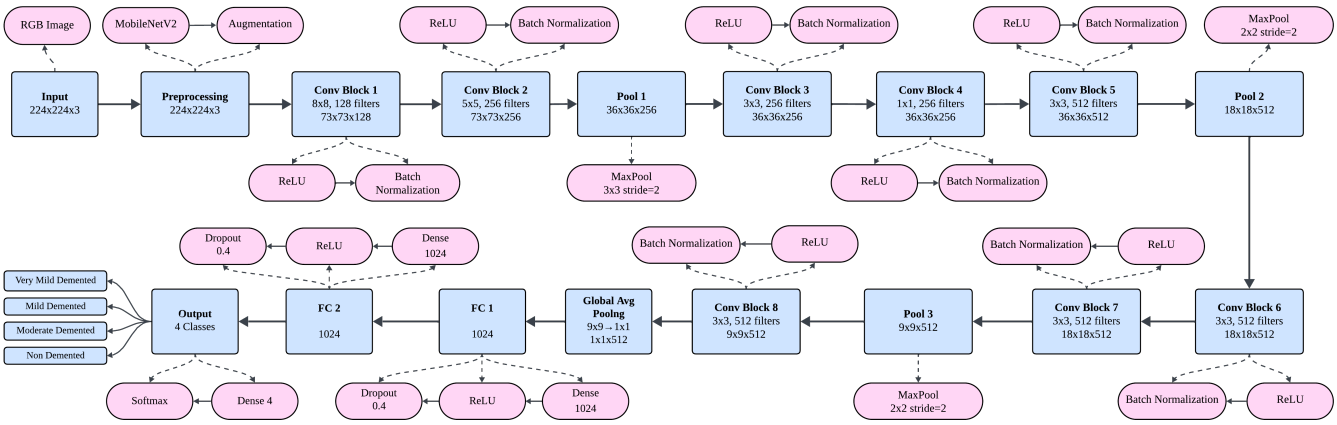


Fig. 1: Proposed Multi-Scale CNN Architecture

The unaugmented dataset was kept completely separate and used as an independent test set for final evaluation, including confusion matrix and ROC analysis. As subject identifiers are not available, splitting was performed at the image level; therefore, subject-level independence cannot be strictly guaranteed. Refer to Table I.

TABLE I: Dataset Description and Evaluation Protocol

Component	**Description**
Augmented Dataset	33,984 images (4 classes)
Unaugmented Dataset	5,099 images (4 classes)
Evaluation Strategy	3-Fold Stratified Cross-Validation (Augmented Set)
Training per Fold	$\approx$ 66.7% of augmented data
Validation per Fold	$\approx$ 33.3% of augmented data
Independent Test Set	Unaugmented dataset (5,099 images)
Performance Reporting	Mean $\pm$ Std (CV) + Independent Test Metrics

C. Data Preprocessing

Pre-training, all the MRI scans were normalized to a universal numeric range and resized to architecture- specific sizes with conversion into three-channel formats to maintain anatomical information. To overcome limited data and biased classification, the training images were subjected to medically reasonable augmentations that comprised rotations (up to 30°), translations (up to 20%), controlled zoom (0.99–1.01 and larger scales), shear, horizontal flips, and brightness changes (0.8–1.2), adding realistic variation with anatomical integrity. The validation and test sets were left unaugmented and only applied normalization and resizing to maintain unbiased testing. For interpretability purposes, the preprocessed images and the activation maps were remapped into displayable ranges and synchronized with the original scans.

IV. Model Architecture

This section describes the architecture of the proposed Multi-Scale CNN model shown in Fig. 1. The model integrates MobileNetV2-based preprocessing, multi-scale convolution blocks, dual-path pooling, and a Global Average Pooling–driven classification head tailored for Alzheimer's MRI classification.

Total parameters: 11,350,404 (43.30 MB)
Trainable parameters: 11,344,516 (43.28 MB)
Non-trainable parameters: 5,888 (23.00 KB)

A. Custom CNN

The proposed architecture is a multi-scale convolutional framework that is specifically optimized for structural MRI rather than the standard CNN. A preprocessing stem based on MobileNetV2 first extracts foundational anatomical edges and intensity gradients, therefore allowing for more stable downstream optimization. Our feature extractor is based on a progressive multi-receptive-field approach. First, an 8x8 convolution with stride 3 is used to capture coarse neuroanatomical structures such as cortical thickness and ventricular enlargement. Then, 5x5 and 3x3 blocks are used in the following stages. Finally, a 1x1 bottleneck convolution is used for further feature compression. Furthermore, a two-stage pooling approach is applied, where mid-scale and high-scale representations are considered. Finally, Global Average Pooling is applied in order to ensure spatial invariance and reduce the number of model parameters. Overall, all of these modules contribute towards better multi-class discrimination while ensuring computational efficiency.

B. Transfer Learning

We employed transfer learning using pre-trained convolution backbones, which can often result in faster convergence and better classification on the limited number of MRIs. The initial layers preserved the overall visual information like edges and textures whereas final layers were fine tuned for Alzheimer's MRI with fewer task-specific data but better generalizability.

1) ResNet50: ResNet50 was utilized because its deep residual architecture allows for the efficient training of very deep networks and the extraction of hierarchical features in an image. Pre-trained layers, using transfer learning, provide low- and mid-level visual representations while higher layers are fine-tuned for the classification of Alzheimer's disease stages. Inputs were resized and normalized to the model's expected format;

a similar preprocessing step was undertaken across models to ensure a fair comparison.

2) EfficientNetB3: In this work, EfficientNetB3 is adopted due to its efficiency of parameters and its powerful ability to transfer learning. EfficientNetB3 has a compound-scaling framework that provides compact yet expressive feature maps. For class-specific pattern extraction from MRIs, fine-tuning of such features was performed. Input preprocessing and data splits are uniformly applied to allow unbiased and repeatable assessment.

3) Xception: Xception was chosen because of depthwise separable convolutions as well as fast feature factorization. Pre-trained filters provided dense mid-as well as high-level descriptors that characterized subtle structure as well as texture variations in MRI scans. Upper layers were fine-tuned under transfer learning to stage Alzheimer's disease while the inputs were normalized as well as resized to make full use of the pre-trained representation.

C. Architectural Novelty

The architectural novelty of the proposed model lies in four components: (1) a MobileNetV2-inspired preprocessing stem for lightweight anatomical feature extraction, (2) multi-scale representations end-to-end directly from MRI images using convolutional receptive fields (8×8 → 5×5 → 3×3 → 1×1), (3) a dual-path pooling mechanism for preserving both mid-level and high-level spatial cues, and (4) a Global Average Pooling–driven classification head that minimizes parameter count while improving generalization. To the best of our knowledge, few prior AD MRI classification frameworks jointly combine these mechanisms into a unified architecture.

V. Results

To evaluate the validity of the outputs provided by our proposed model, we conducted many experiments on our dataset. The experimental findings and an analysis of our suggested model are presented in this section.

A. Cross-Validation Performance

The accuracy for the validation set, as shown in Fig.2, increased monotonically over the course of 30 epochs. There were no signs of overfitting evident in the data for one representative fold of the 3-fold cross-validation. For the entire set of folds, the mean accuracy for the new model was 99.15%, with a standard deviation of 0.04. The mean F1-score for the new model was 0.9915 with a standard deviation of 0.0004.

B. Confusion Matrix

The confusion matrix in Fig. 3 presents the class-wise prediction distribution across the four Alzheimer's disease stages on the independent unaugmented test set. The model achieved high classification accuracy with only a few misclassifications (e.g., 2 Non-Demented samples predicted as Mild Demented and 5 as Very Mild Demented). Overall, the results indicate strong multi-class discrimination performance.

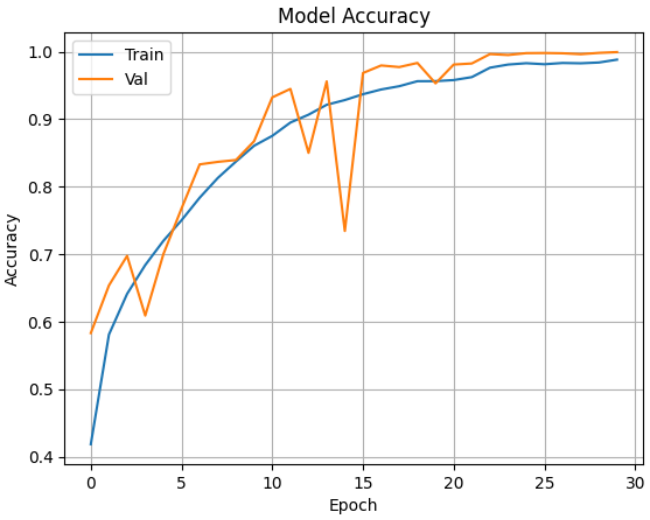


Fig. 2: Training vs Validation Accuracy

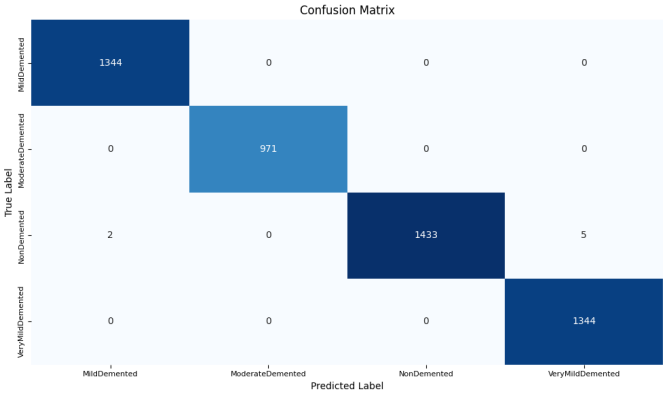


Fig. 3: Confusion Matrix

C. Performance of Individual Models

The proposed Custom CNN was compared to EfficientNetB3, ResNet50 and Xception with similar training settings. As is observed from Table II, it achieved the highest cross-validation accuracy while consuming reasonable number of parameters as well as inference time. Xception and ResNet50 also performed exceptionally, while EfficientNetB3 achieved lesser accuracy but with efficient parameters.

TABLE II: Performance Comparison of Individual Models

Model	Accuracy	F1	Inf. Time(s)
CNN (Proposed)	**99.15%**	**0.9915**	**15.2**
ResNet50	98.31%	0.9832	22.8
Xception	98.22%	0.9825	20.1
EfficientNetB3	96.15%	0.9618	18.5

As summarized in Table II, the proposed Custom CNN outperforms the other models in cross-validation accuracy and F1 score while maintaining competitive inference time for test set.

D. Explainable AI

1) Grad-CAM Guided Backpropagation Overlay: Guided Backpropagation with Grad-CAM produces high-resolution class-discriminative heatmaps from low-resolution input MRI volumes that localise regions important to discriminating Alzheimer's stages and verifying that the network is looking at relevant structure instead of artifacts, as seen in Fig. 4.

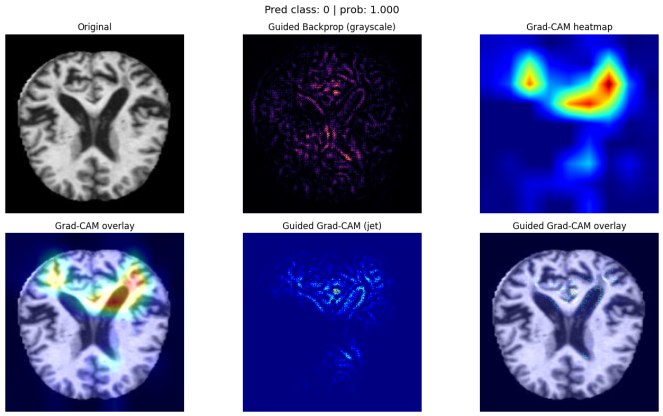


Fig. 4: Grad-CAM Guided Backpropagation Overlay for Mild Demented

2) Integrated Gradients: Integrated Gradients assigns pixel-wise importance relative to a baseline, producing overlays that highlight regions most influential for classification. These visualizations support interpretability by indicating a focus on anatomically plausible regions rather than imaging artifacts; however, quantitative XAI validation is left for future work. See Fig. 5).

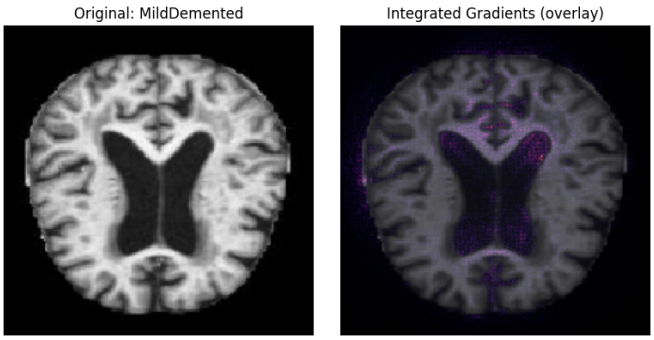


Fig. 5: Integrated Gradients for Mild Demented

3) LIME: LIME generates superpixel-based heatmaps highlighting regions used by the model; qualitatively, these overlays correspond to anatomically plausible structures and enhance interpretability. While the visualizations may support clinician interpretation, quantitative XAI validation and external assessment are left to future work as shown in Fig. 6.

E. ROC Curve

The high discriminatory ability of the model is also evident from the ROC curve plots for all the four classes, as presented

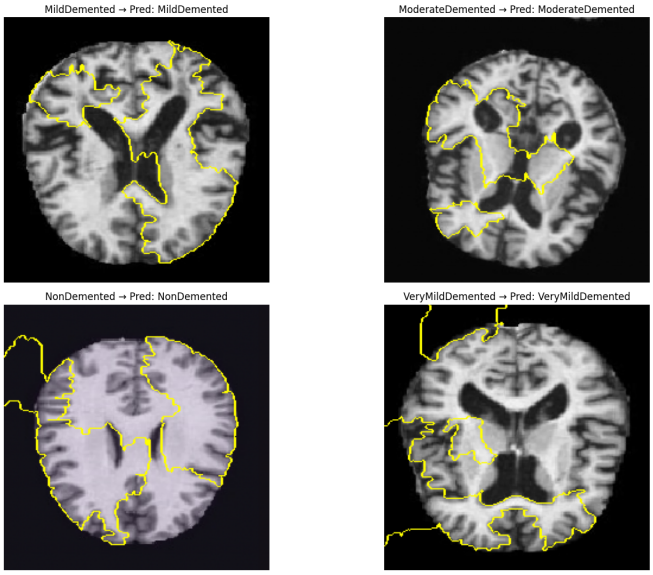


Fig. 6: LIME Visualization

in Fig. 7. High AUC values were obtained for all the classes, which indicates the good performance of the model in class separation. This is in agreement with the cross-validation accuracy and F1-score results, which also point to the good discriminatory ability of the model on the dataset, as presented in Table III.

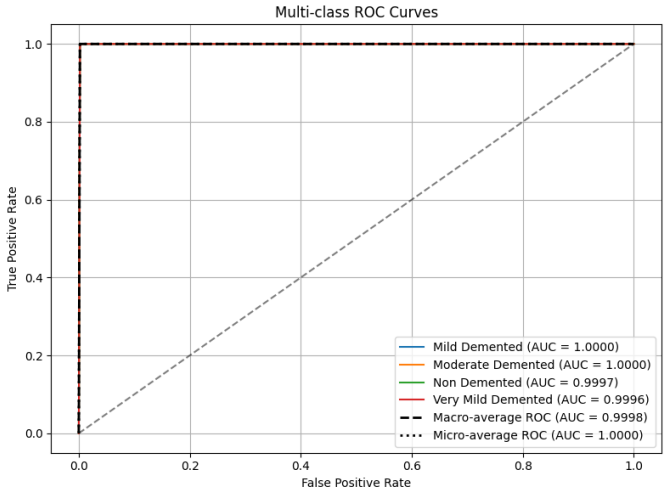


Fig. 7: ROC curve of Proposed Model

TABLE III: Performance Metrics for Alzheimer's Disease Classification

Class	Precision	Recall	F1-Score	AUC
Mild Demented	0.9929	0.9977	0.9953	1.0000
Moderate Demented	1.0000	1.0000	1.0000	1.0000
Non Demented	0.9938	0.9820	0.9878	0.9997
Very Mild Demented	0.9817	0.9895	0.9856	0.9996
Macro Avg	0.9921	0.9923	0.9922	0.9998

F. Web Application

For ease of access and usability of our research, the proposed models were deployed in a Streamlit-based web application [19]. The users can choose any one of the four trained models: Custom CNN, Xception, ResNet50, or an ensemble that combines the three using majority voting. Such an ensemble enhances robustness in prediction without compromising computational efficiency.

VI. Comparative Analysis

Table IV summarizes the comparative analysis, showing that many existing studies focus on binary classification or report lower multiclass accuracy, while most high-performing methods do not jointly address interpretability, explainability techniques and deployment feasibility.

TABLE IV: Comparison of Different Studies

Study	Classes	Accuracy	Explainability	Deployment
[3]	2	99.05%	No	No
[4]	2	98%	No	No
[5]	4	100%	No	No
[6]	2	90.3%	Grad-CAM	No
[8]	2	94.65%	No	No
[9]	2	98.74%	No	No
[10]	2	95%	No	No
[11]	2	91.35%	No	No
[12]	4	95.23%	No	No
[13]	2	98%	No	No
[14]	4	99.65%	No	No
[15]	4	96.6%	Grad-CAM	No
[16]	4	98.8%	No	No
[17]	4	99.74%	No	No
[18]	4	95.67%	No	No
Proposed	4	**99.15%**	Grad-CAM, LIME, Integrated Gradients	**Yes**

VII. Conclusion and Future Work

Although excellent results were obtained in each class, cross-validation of the enhanced dataset yielded consistent results, and testing of the unaugmented dataset verified that generalization is occurring within this dataset. The proposed deep learning system achieved reliable multi-class Alzheimer's disease classification from structural MRI scans. The custom CNN model obtained an average accuracy of 99.15% $\pm$ 0.04 and an F1-score of 0.9915 $\pm$ 0.0004, with computational complexity comparable to Xception, EfficientNetB3, and ResNet50. The ensemble of the three models further improved prediction stability in borderline cases.

The explainability techniques Grad-CAM-based backpropagation, Integrated Gradients, and LIME were improved. Cross-validation results showed the stability of the models' class-wise performance with the augmented data set in terms of the standard evaluation metrics. For the models to be easily accessed and used, the models were deployed as a lightweight Streamlit web application. The source code is publicly available [20]. For future work, the models would be validated at the subject level with data sets that have patient identifiers, clinically validated in multiple centers, multimodal data would be incorporated, optimization for real-time deployment would be performed, and privacy-preserving learning would be achieved.

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